CHAPTER 10

PRINTING

In this chapter you'll learn about

- Setting Up Your Printer
- Changing Printers and Fonts
- Printing a Standard Business Envelope
- Using Laser Printers and Other Page Printers

This chapter presents some basic information about printing in WordStar.

SETTING UP YOUR PRINTER

There are two types of ports—parallel and serial—for connecting your printer to your computer. With a parallel port connection, the printer receives all eight bits of character codes at the same instant over eight wires. With a serial port, bits are sent one at a time over fewer wires. Printers are connected by cables to the ports located at the back of the computer. When you install a printer in WordStar, you need to specify which port your printer is connected to. LPT ports are parallel ports and COM ports are serial ports. The default port in WordStar is LPT1.

If you have a serial connection, make sure your AUTOEXEC.BAT file contains the correct mode statement. The mode statement should look like this:

mode com1:9600,n,8,1,p

If necessary, replace "com1" with the correct port number for your system, and replace "9600" with the baud rate your printer is set for. For more information, see the manual that came with your printer.

CHANGING PRINTERS AND FONTS

When you install a printer, you specify the name of your printer, the fonts to use, and the computer port. WordStar creates a printer description file (PDF) with this information. You assign a name to the PDF; for example, if you have a laser printer, you might call it LASER. WordStar adds the extension .PDF, so the printer description filename would be LASER.PDF. When you create a document, the PDF for your default printer is assigned to that document. For more information about installing printers, see the *Installing and Customizing* booklet.

CHANGING PRINTERS

If you installed more than one printer to work with WordStar, and you want to print with a different printer, you can change the printer assigned to a document.

To use a different printer than the one assigned to your document, follow these steps:

1. Open a file.
2. Choose File→Change Printer (Alt+F,H).

```
Change Printer

Printer for This Document: HP3 ↓                    OK
                                                    Cancel

Printers:
ASC256  | ASCII   | DRAFT   | HPLASER | WS4     |
```

In the Change Printer dialog box, WordStar displays the name of the printer assigned to this document and a list of the installed printers.

3 Select a printer from the list.

4 Press F10 or choose OK to close the dialog box.

CHANGING FONTS

You can display a list of fonts that are available for the current printer. You can then change fonts by following these steps:

1 Choose Style→Font (Alt+S,F).

```
Font

Font: Courier 10 PC ↓                               OK
                                                    Cancel

Fonts:              Printer: HPLASER
Courier 10 PC     | Courier 12 PC   | Line Pr 17 PC   | Times PC ...
Univers PC ...    |
```

You see the name of the current printer as well as the font now in effect. The other available fonts are listed.

2 Select one of the fonts and close the dialog box.

The font tag appears in the text area. Text will reformat in the paragraph containing the font tag. Subsequent paragraphs affected by the new font need to be reformatted.

3 Choose Utilities→Reformat→Rest of Document (Alt+U,R,R). The rest of the paragraphs are reformatted. The font tag is in effect until another font tag is inserted.

TIP You can also use the classic command ^QU to reformat text.

Follow these steps to see how the document will look when printed:

1 Choose View→Preview (Alt+V,P) to view the page with the new fonts.

2 Press Esc to exit Preview.

3 Close and abandon the file.

For more information about fonts, see "Using Fonts With WordStar" in this chapter.

PRINTING A STANDARD BUSINESS ENVELOPE

You can print envelopes by setting a shorter page length and special margins. The format in the example below is suitable for a standard 9.5 x 4-inch business envelope, printed 10 characters per inch. It is designed for printers such as dot matrix and daisy wheel printers that are nonlaser and nonpage printers.

To create a standard format for envelopes, follow these steps:

1 Open a new file.

2 Choose Layout→Page (Alt+L,P) to see the Page Layout dialog box.

3 Specify the following settings for the envelopes:

Odd page offset	4.00"
Even page offset	4.00"
Top margin	2.00"
Bottom margin	0.00"
Page length	4.00"

4 Choose Layout→Page Numbering→Print Page Numbers (Alt+L,N,P) to omit page numbers from the envelope. Press F10 or choose OK.

5 Save and name the envelope format file.

If you use different sizes of envelopes, you can create individual formats using different margin and tab settings. You can store each format in a separate file. Then, when you want to print an envelope, you insert the appropriate format file following the last page of your letter.

Follow these steps to insert an envelope format file so you can print the envelope immediately after you print a letter:

1 Open an existing file (letter) or use the sample file MAILING.DOC.

2 Choose Edit→Go to Other→Document End (Alt+E,O,O) to go to the end of the file.

If your file has headers or footers, complete step 3. If not, move to step 4. (If you're using MAILING.DOC which has no headers or footers, you can move to step 4.)

3 Choose Layout→Headers/Footers (Alt+L,H). Select the same header or footer commands that exist in your letter.

You add no text to the new headers and footers and they cancel your previous headers or footers. You don't want the headers or footers to print on the envelope.

4 With the cursor at the end of the file, choose Insert→Page Break (Alt+I,P) to insert a page break. Your envelope format will follow the page break.

5 Just below the page break, choose Insert→File (Alt+I,F).

6 Type the name of the envelope file that you created in the previous section. Press F10 or choose OK.

The envelope file is inserted at the end of the document.

7 Beginning in line 1, column 1, immediately below the dot commands for the envelope, type the name and address for the envelope.

8 Save the file.

9 When you print the file, select the print option Pause Between Pages so you can insert an envelope in the printer at the appropriate time.

If you have a laser printer, WordStar provides special envelope files. HP-ENV.LST is for the HP LaserJet, the LaserJet Plus, and other printers that feed envelopes from the right side of the printer tray. HP2-ENV.LST is for the HP LaserJet series II and III and other printers that feed envelopes from the center of the manual feed tray. For more information on these files, see the "MailList" section of the *Companion Programs* manual.

ABOUT LASER PRINTERS AND OTHER PAGE PRINTERS

Page printers print a page all at once rather than one character or line at a time as nonpage printers do. Laser printers are page printers. Following are some terms that may be useful to know if you have a page printer:

PROPORTIONAL AND NONPROPORTIONAL FONTS

A nonproportional font has fixed spacing, that is, every letter takes up the same amount of space. A proportional font has proportional spacing, that is, some characters are wider and take up more space than others. Text printed with a nonproportional font has an inconsistent appearance, with gaps between some of the letters. Text printed with a proportional font is spaced more consistently on the page. Some examples of the two types of fonts are shown below.

NONPROPORTIONAL	PROPORTIONAL
mmmmmmmm	mmmmmmmm
iiiiiiii	iiiiiiii

TYPE SIZES

The size of a font is measured in points, for example, Times Roman 10-point. There are approximately 72 points in an inch. Font height is measured from the lowest *descender* (for example, the bottom of the letter "p") to the highest *ascender* (the top of a capital letter) including a small amount of extra space.

For long sections of text 9-, 10-, and 11-point fonts are the easiest to read.

SERIF AND SANS SERIF

Serifs are the small lines that cross the main strokes of a letter. An example of a serif typeface is **Times Roman**; an example of a sans (without) serif typeface is **Helvetica**. Note that Helvetica has only the strokes required to define the letter, while Times-Roman has extra little lines.

Most typographers feel that serif type is easier to read in long paragraphs of text because there are more lines defining each letter. Sans serif type is used for headings or short paragraphs. Because sans serif type looks more modern, it is often used for technical documents.

TYPE FAMILIES AND TYPEFACES

A *type family, font family,* or *typeface* is a type design, like Times Roman or Helvetica. It's best to limit yourself to two or three type families per document. Be careful when you mix type families on a page. Some typefaces look better together than others. Generally speaking, a serif and sans serif typeface or two sans serif typefaces look better on a page than two serif typefaces.

Each type family contains several variations within it, for example, roman (upright, regular weight), **bold**, and *italic*. Bold and italic are difficult to read in long sections of text, but are good for adding emphasis to particular words or phrases.

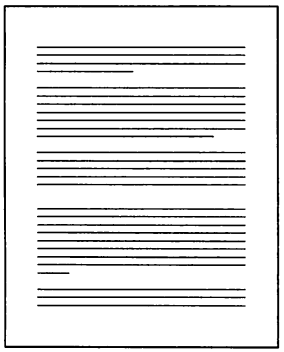
SYMBOL SET

A symbol set is a set of characters that a font contains but that may not appear on your keyboard. The actual characters available depend on your computer and your printer. A character must be included in the symbol set of the font you're using or it won't print. Symbol sets may contain international characters or line and box drawing characters.

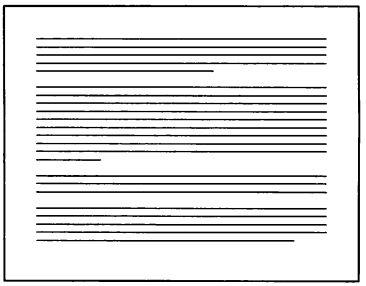
Orientation

Orientation refers to the position of text on the page. Portrait-mode orientation prints text across the width of the page; landscape-mode orientation prints text across the length of the page. Landscape orientation is available only with page printers.

Using fonts with WordStar



portrait



landscape

Many fonts are available for page printers. Fonts fall into three categories: internal, cartridge, and soft.

- **Internal fonts** Every page printer has at least one internal font. Information about internal fonts is stored permanently in the printer's memory. Most of these fonts are nonproportional, portrait-mode fonts. They are always available to the printer.

  Unless you replace the internal fonts with other choices (when you install your printer), these fonts are available when you press **^P=** at the WordStar editing screen.

- **Cartridge fonts** You can purchase cartridges that contain extra fonts. To use a cartridge font, you must insert the appropriate cartridge into the printer.

  To use a cartridge font with WordStar, you specify the fonts on that cartridge when you install WordStar. If you purchase the cartridge later, you use the PRCHANGE option in WINSTALL to install the cartridge. If the cartridge fonts are HP LaserJet scalable fonts, use the LSRFONTS option.

- **Soft fonts** Soft fonts are contained on floppy disks. To use these fonts, you must download them, or send the font information from the disk to the printer.

  You include soft font choices in your WordStar printer description file (PDF). You add soft fonts to a PDF using the PRCHANGE option. Some soft fonts are supplied with WordStar. You can install these fonts when you install your printer. For soft fonts that aren't listed in PRCHANGE, use the LSRFONTS option to add the fonts to the customized database. Then use PRCHANGE to add the fonts to a PDF. The fonts must be downloaded to the printer.

  For more information on downloading fonts, see the *Installing and Customizing* booklet.

BITMAPPED AND SCALABLE FONTS

All fonts are either bitmapped or scalable.

- **Bitmapped fonts** Bitmapped fonts hold the characteristics for the size of each letter in an individual bitmap, or blueprint, which indicates to the printer where to place every dot of ink. Because printers store bitmapped fonts in memory, they are ready to use quickly and require a minimum of software instructions.

- **Scalable fonts** Scalable fonts are sets of instructions that tell a utility, known as a *font generator*, how to create a specific bitmapped font. Because one scalable font can generate a variety of point sizes, type widths, or even italic, a scalable font is an economical option for someone who needs a large selection of fonts.